e-waste umbrella—95 items compared to 21 earlier, from toys to iron boxes. Next, in a departure from the current practice, the Central Pollution Control Board and not the state boards will award licences to recyclers. Finally, a mechanism for an online portal for transactions, including EPR certificates that producers of goods can buy, has been proposed. "Overall, these regulations are extremely positive and a step in the right direction," says Nitin Gupta of Attero Recycling, which is among India's largest e-waste processing companies. Based in Roorkee, Uttarakhand, the plant is basically a 200-metre-long corridor with three sheds one after another. The first shed houses administrative offices in front and the primary metal shredding unit at the back. Large e-waste items such as ACs, refrigerators, large TV panels, metal sheets, etc. are shredded and sorted in smaller bits for recycling. The second shed has thousands of lithium-ion batteries—mostly from automobiles— stacked in a neat pile, while workers dismantle their components one by one. In the rear are micro component separation areas, where smaller and specialised e-waste is broken up.

An upper floor in this shed has an army of workers dismantling and extracting minerals out of e-waste from smaller items such as phones, printers, laptops and smaller LCDs. This floor also has material testing labs as well as a section for extracting precious minerals. The far end of the shed has micro platforms of hydro metallurgical processes to extract metals and minerals from soluble e-waste. The last shed houses the furnaces where metals like copper and aluminium are smelted from segregated waste and poured into moulds to form ingots. Large water troughs on the opposite side have separate soluble waste using silting and ion exchange procedures. At the forefront of e-waste recycling in India, Attero processes about 160 tonnes of e-waste every day. Building on its success, the company is preparing to open two more plants—one in India and another in Poland.

Not everyone in the current supply chain, though, is as optimistic as Attero's Gupta. "There is a lot of worry about these rules," says Pranshu Singhal, founder of Karo Sambhav, a PRO with 55 collection centres across the country that pick up 5,000 tonnes of e-waste annually. The new draft disarms the PROs of that mandated role, Singhal fears. Though the proposed rules do not mention PROs or dismantlers specifically, they reduce the number of recognised entities in the supply chain to four—manufacturer, producer, refurbisher and recycler. "There is very little focus on the setting up of collection channels which, for me, is at the heart of going circular. We are imagining that the recycler will magically collect (e-waste)," says Singhal. Shobha Raghavan, COO of the Bengaluru-based non-profit Saahas Zero Waste, one of the early entrants in waste collection, agrees. It's only in the past five years when e-waste rules kicked in that the whole sector began moving towards a more formal system involving massive awareness and training outreach. "Disrupting an ecosystem built over five years by removing the PRO and dismantler entities was unexpected," she says. "Nowhere in the world does an industry have a supply chain with just a starting and an ending point."

THE WAY FORWARD

While the new draft rules do need to accommodate such concerns, there's now a simultaneous focus on sustainable consumption. In July, the Union ministry of consumer affairs set up a committee to work on a comprehensive Right to Repair legislation framework, the rationale being that consumers should be able to repair and modify their products. "Right to repair has a very broad ambit, which also includes the problem of e-waste," says Prof. Ashok Patil, chair of Consumer Law and Practice at the National Law School of India University, who is on the committee. The situation has reached dire proportions. Consider the uptick in smartphone and laptop sales during the pandemic, which necessitated working from home and online classes. These devices will reach the end of life in 4-5 years and enter the scrap market. Are we ready for it? "Obviously not," says Ravi Kumar Neeladri, CEO of Cerebra Green, a Bengaluru-based refurbisher and recycler. "The steps are all in the right direction, but it all depends on how we integrate the informal segment into the formal," he says. "It is a sunrise segment in my opinion and we have not even scratched the surface."

> Experts stress the need to **push manufacturers to make products** that have longer life-spans and use components that can be safely recycled

Experts also propose long-term solutions to manage e-waste, including pushing manufacturers to develop environment-friendly products that not only have longer life spans but also use material that can be safely recycled. Priti Mahesh of Toxics Link advocates a rethink on the way products are designed—for longevity, as it used to be, and not for the dump, as it is now. "The life span of products has obviously come down in the past 5-7 years." The responsibility rests with us too. As consumers, we should insist that all products we buy can be recycled safely and not end up in electronic graveyards, returning to haunt us as toxic waste. ■

–with Anilesh S. Mahajan and Kiran D. Tare